

Minimal Kinetic Model of Direct Air Capture of CO₂ by Supported Amine Sorbents in Dry and Humid Conditions

Hongjun Liu, Hongfei Lin, Sheng Dai, and De-en Jiang*



Cite This: *Ind. Eng. Chem. Res.* 2024, 63, 5871–5879

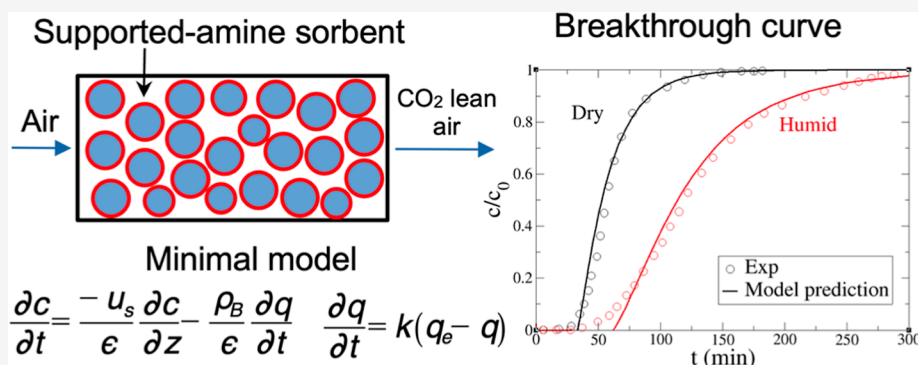


Read Online

ACCESS |

Metrics & More

Article Recommendations



ABSTRACT: Dilute concentration (~400 ppm) and humidity are two important factors in the direct air capture (DAC) of CO₂ by supported sorbents. In this work, a minimal DAC CO₂ adsorption-kinetics model was formulated for supported amine sorbents under dry and humid conditions. Our model fits well with a recent DAC experiment with supported amine sorbent in both dry and humid conditions. Temperature and flow rate effects on breakthrough curves were quantitatively captured, and increasing temperature led to faster CO₂ adsorption kinetics. Moisture was shown to broaden the breakthrough curve with slower CO₂ adsorption kinetics but significantly improve the uptake capacity. The present minimal model provides a versatile platform for kinetic modeling of the DAC of CO₂ on supported amine and other chemisorption systems.

1. INTRODUCTION

Direct air capture (DAC) of CO₂ is an important solution to control CO₂ concentration in the atmosphere in order to mitigate climate change due to the increasing CO₂ level.^{1–3} The DAC process usually involves the use of liquid or solid sorbents to selectively capture CO₂ from the air via chemisorption, followed by sorbent regeneration via a temperate/pressure swing to release concentrated CO₂ for storage or utilization. Supported amine sorbents have attracted significant attention for DAC, owing to their high adsorption capacity, excellent selectivity, beneficial coadsorption with moisture, and relatively mild energy requirements for sorbent regeneration.^{4–10} Physically impregnated or chemically grafted amine groups onto porous supports such as mesoporous silica, alumina, carbon, and metal–organic frameworks have been reported.^{11–14} The capture capacity is dictated by the reaction stoichiometry of amine groups with CO₂ in dry or humid conditions, the temperature, and the loading of amines in the porous supports.^{9,15–18}

The dilute concentration of CO₂ in ambient air at about 400 ppm is one major challenge for DAC, requiring a strong thermodynamic driving force for its sorption. Most of the experimental and modeling work on amines focused on much

higher CO₂ partial pressures.^{6,19–22} Enhanced CO₂ adsorption in the presence of H₂O has been widely observed in amine adsorbents.^{2,23–25} Moreover, other chemisorption systems using superbase-derived ionic liquids have been developed.^{26,27} Although researchers have turned their attention to DAC conditions,^{8,28,29} how sorbent properties and H₂O affect CO₂ adsorption at DAC conditions is not well understood from a modeling perspective.^{30–32} On the experimental side, Sayari,³³ Jones,¹ and many others have extensively examined amine-functionalized mesoporous silica for DAC. The effect of water on CO₂ adsorption has been recently reviewed by Sayari and co-workers,¹⁰ while Jones and co-workers have investigated many different types of porous supports loaded with poly(ethylenimine) (PEI).^{34,35} Molecular simulations at both the molecular mechanical and quantum mechanical levels have

Received: December 20, 2023

Revised: March 4, 2024

Accepted: March 5, 2024

Published: March 19, 2024



been employed to understand the DAC of CO₂ and its interaction and reaction with amines as well.^{36–39}

A few recent kinetic modeling efforts have examined carbon capture under the DAC conditions. Stampi-Bombelli et al. designed and analyzed temperature-vacuum swing adsorption cycles for DAC.³¹ Elfving and Sainio developed a detailed and new kinetic approach to model CO₂ adsorption from humid air in amine-functionalized resins based on their experimental measurements.³² Young et al. proposed two different models to consistently describe H₂O and CO₂ coadsorption and to investigate their effects on DAC with amine-functionalized polymer-bead sorbents; they further performed process optimization for the DAC cycles.⁴⁰ Overall, these very recent models are geared toward DAC process optimization, but for many researchers who are discovering and testing new porous sorbents at the benchtop scale, a simpler kinetic model that can fit the measured breakthrough curves well at various flow, temperature, and humid conditions would be more desirable to quickly evaluate the key thermodynamic and kinetic factors.

In this work, we develop a simple kinetic model of the DAC of CO₂ by supported amine adsorbents to examine the effects of relevant parameters on breakthrough curves in dry and humid conditions in a packed bed and to help researchers quickly identify important thermodynamic and kinetic factors that can be used for further improvements of the DAC performances of amine-functionalized sorbents and other chemisorption systems. Below, we first describe our modeling approach and considerations and then validate our minimal model by comparing it to the literature. Next, we performed a parametric study to explore the effect of different parameters on breakthroughs in dry and humid conditions. Finally, we model a recent experimental DAC work to provide physical insights into the adsorption kinetics.

2. METHODS

In the DAC process, a trace level of CO₂ is adsorbed from a nonadsorbing carrier so that a single component model applies under dry conditions and a binary model under humid conditions. The mass transfer between gas and solid phases includes three resistances (external film, macropore, and micropore) and can be depicted by a linear driving force (LDF) model, where a lumped uptake rate constant is used to take various resistances into account.⁴¹ The LDF model is conceptually simple, computationally efficient, and therefore widely used in the literature.^{21,42} Momentum balance in the mobile gas phase includes viscous and kinetic terms; in a lab scale setup, constant gas velocity and negligible pressure drop can be assumed due to the short column length and low flow rate. Energy balance consists of three heat transfer equations for the gas phase, solid phase, and column wall. Local thermal equilibrium is assumed between gas and solid phases and also between the wall and ambient environment so that only one energy balance is needed for the whole packed bed system.

The mass balance equation in the gas phase is given as follows

$$\frac{\partial c}{\partial t} = D_L \frac{\partial^2 c}{\partial z^2} - \frac{u_s \partial c}{\partial z} - \frac{\rho_B \partial q}{\epsilon \partial t} \quad (1)$$

where the three terms on the right-hand side are dispersion, convection, and adsorption on supported adsorbents, respectively; c is the gas concentration (mol/m³), D_L the axial dispersion coefficient (m²/s), u_s the superficial velocity (m/s),

ρ_B the adsorbent bulk density (kg/m³), ϵ the bed void fraction (dimensionless), and q the adsorbate concentration (mol/kg). The LDF model (eq 2) was used to describe adsorption kinetics

$$\frac{\partial q}{\partial t} = k(q_e - q) \quad (2)$$

where k (s⁻¹) is the uptake rate constant (determined by optimizing the solution to the LDF model against the experimental breakthrough) and q_e (mol/kg) is the equilibrium adsorption concentration. q_e was determined by fitting the equilibrium experimental isotherms to the Toth adsorption model (eq 3) with temperature-dependent parameters b (atm⁻¹), n_s (mol/kg), and t_T (dimensionless) expressed in eqs 4–6, respectively, where n_{s0} (saturation adsorption), b_0 (adsorption affinity), ΔH_0 (J/mol, heat of adsorption), and t_{T0} (Toth exponent) were optimized through nonlinear fitting to the experimental isotherm data. The amine-loading in the sorbent was not explicitly described in our model; instead, it was captured indirectly by the Toth parameters.

$$q_e = \frac{n_s b p}{(1 + (b p)^{t_T})^{1/t_T}} \quad (3)$$

$$b = b_0 \exp\left(\frac{\Delta H_0}{RT_0} \left(\frac{T_0}{T} - 1\right)\right) \quad (4)$$

$$n_s = n_{s0} \exp\left[\chi \left(1 - \frac{T}{T_0}\right)\right] \quad (5)$$

$$t_T = t_{T0} + \alpha \left(1 - \frac{T}{T_0}\right) \quad (6)$$

Similarly, the heat balance equation (eq 7) consists of dispersion, convection, adsorption, and wall heat transfer terms

$$\frac{\partial T}{\partial t} = \frac{1}{\epsilon \rho_g C_p^g + \rho_B C_p^s} \left[K_L \frac{\partial^2 T}{\partial z^2} - \rho_g C_p^g u_s \frac{\partial T}{\partial z} + \rho_B \Delta H \frac{\partial q}{\partial t} - \frac{2h}{R} (T - T_w) \right] \quad (7)$$

where C_p is the specific heat capacity (J kg⁻¹ K⁻¹), K_L axial thermal conductivity (W m⁻¹ K⁻¹), h wall heat transfer coefficient (W m⁻² K⁻¹), and R column radius (m). Initial conditions used were $c = 0$, $q = 0$, and $T = T_0$ at $t = 0$, and boundary conditions were $c = c_0$ and $T = T_0$ at $z = 0$, and $\partial c / \partial z = 0$ and $\partial T / \partial z = 0$ at $z = L$.

The Guggenheim-Anderson-de Boer (GAB) model was used to describe H₂O uptake (q_2) for both single component and binary systems as a function of relative humidity (ϕ , dimensionless)

$$q_2^e = \frac{c_m c_G K_{ads} \phi}{(1 - K_{ads} \phi)[1 + (c_G - 1)K_{ads} \phi]} \quad (8)$$

where c_m (mol/kg), c_G (dimensionless), and K_{ads} (dimensionless) are model parameters. The saturated vapor pressure can be calculated by the empirical Buck equation

$$p^{\text{sat}}(T) = 0.61121 \exp\left[\left(18.678 - \frac{T}{234.5}\right)\left(\frac{T}{257.14 + T}\right)\right] \quad (9)$$

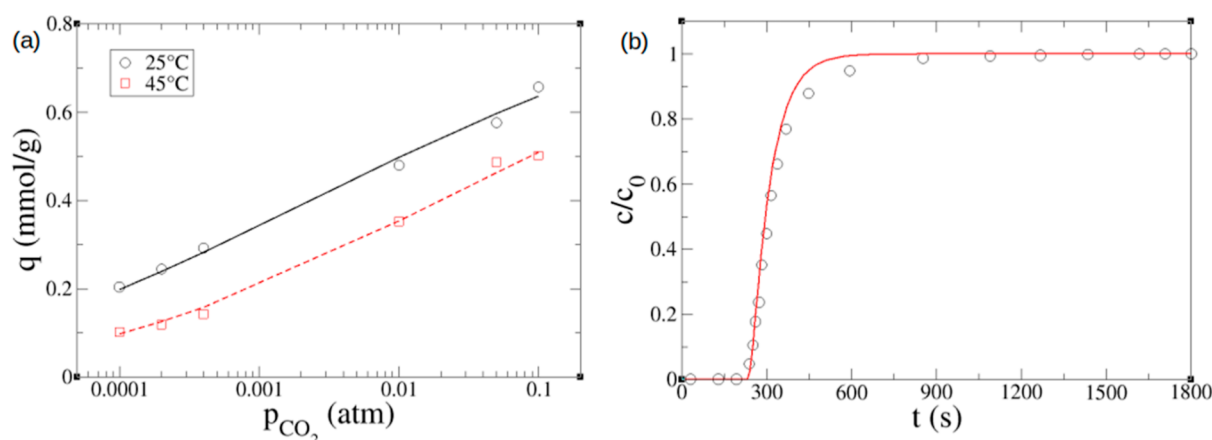


Figure 1. (a) Toth model fit (line) using eq 3 to the experimental isotherm data (symbol) in amine-functionalized SBA-15 adsorbent (via 3-aminopropylsilyl groups) at two different temperatures (experimental isotherms were measured with a thermogravimetric analyzer for sorbent pellets of 150–550 μm in size).²¹ (b) Comparison between model prediction (line) and experiment (symbol) of breakthrough profiles of the amine-functionalized SBA-15 sorbent (0.1 atm CO_2 , 25 $^\circ\text{C}$, dry).²¹

where p^{sat} is in the unit of kPa and T of $^\circ\text{C}$.

Method of lines was applied to solve PDEs numerically. N grid points were discretized on a length scale. The second-order centered difference approximation formula was used to estimate the first and second derivatives. Each PDE was subsequently converted to N ODEs. The resulting ODEs were simultaneously solved by the ode15s solver in MATLAB to obtain the profiles of the gas concentration $c(r,t)$, adsorbed gas concentration in adsorbents $q(r,t)$, and gas temperature $T(r,t)$. The ode15s solver uses the variable order method to solve ODEs, which dynamically chooses discretization methods of different orders and sets the time grid automatically; it also uses the variable step-size method.

3. RESULTS AND DISCUSSION

3.1. Model Benchmark at the Flue-Gas Conditions. To validate our implementations, we chose to reproduce the breakthrough curves reported in the amine-functionalized SBA-15 silica adsorbent for the flue-gas conditions (0.1 atm CO_2).²¹ The experimental equilibrium adsorption data were fitted to a Toth isotherm, and the quality of the fitting is very good (Figure 1a). Fitting parameters ($n_s = 0.97$ mmol/g, $b_0 = 8.3 \times 10^5$ atm $^{-1}$, $\Delta H_0 = 84$ kJ/mol, and $t_T = 0.21$) are comparable to the previously reported values ($n_s = 1$ mmol/g, $b_0 = 3 \times 10^5$ atm $^{-1}$, $\Delta H_0 = 75$ kJ/mol, and $t_T = 0.22$).²¹ More importantly, excellent agreement was achieved between our model prediction and the experimental breakthrough profile under the flue gas conditions (Figure 1b). In addition, it has been shown that simple isothermal models provide an accurate description of the mass transport behavior in the adsorption column at the benchtop scale,²¹ so here, we also adopt an isothermal model to study the breakthrough curves at even lower CO_2 concentrations such as DAC conditions (400 ppm of CO_2). The mass balance equation was further simplified by dropping out the second derivative dispersion term, since the convection term is dominant. The LDF kinetic model and Toth isotherm were used to complement the following PDE for the numerical solution. This simplified version (eq 10) is the minimal model for DAC CO_2 adsorption.

$$\frac{\partial c}{\partial t} = \frac{-u_s \partial c}{\epsilon \partial z} - \frac{\rho_B \partial q}{\epsilon \partial t} \quad (10)$$

3.2. Parametric Study at DAC Conditions. Using our minimal kinetic model, we examined the effects of k , u_s , and L on the breakthrough curves at DAC conditions. Uptake rate constant k , a lumped term combining mass diffusion rate and CO_2 adsorption rate, dictates the breakthrough curve shape.⁴³ Superficial velocity u_s and column length L are important controlling parameters for unit operation of the packed adsorption bed. The Toth isotherm and modeling parameters used in the simulations are listed in Tables 1 and 2,

Table 1. Toth and Modified Toth Isotherm Parameters for CO_2 and GAB Isotherm Parameters for H_2O Used in the Parametric Study at DAC Conditions

	parameters	values
Toth for CO_2	b	1.0×10^5 atm $^{-1}$
	n_s	1.0 mmol/g
	t_T	0.25
modified Toth for CO_2 with H_2O	γ	6.0×10^{-3} g/mmol
	β	25 g/mmol
GAB for H_2O	c_m	35 mmol/g
	c_G	0.15
	K_{ads}	0.51

respectively. Several qualitative trends can be clearly identified. The larger uptake rate constant k leads to a steeper breakthrough (Figure 2a) due to the faster adsorption kinetics. Larger gas velocity u_s leads to faster and steeper breakthroughs

Table 2. Preset Input Conditions Used in Parametric Study of Kinetic Modeling at DAC Conditions

parameters	values
bed void fraction ϵ	0.5
adsorbent bulk density ρ_B	0.5 g/cm 3
bed length L	2 cm
superficial velocity u_s	0.8 cm/s
CO_2 uptake rate constant k_1	0.02 s $^{-1}$
H_2O uptake rate constant k_2	0.03 s $^{-1}$
CO_2 feed concentration c_{10} of 400 ppm	1.6×10^{-5} mmol/cm 3
relative humidity ϕ	20–50%
H_2O feed concentration c_{20} at $\phi = 40\%$	2.6×10^{-4} mmol/cm 3

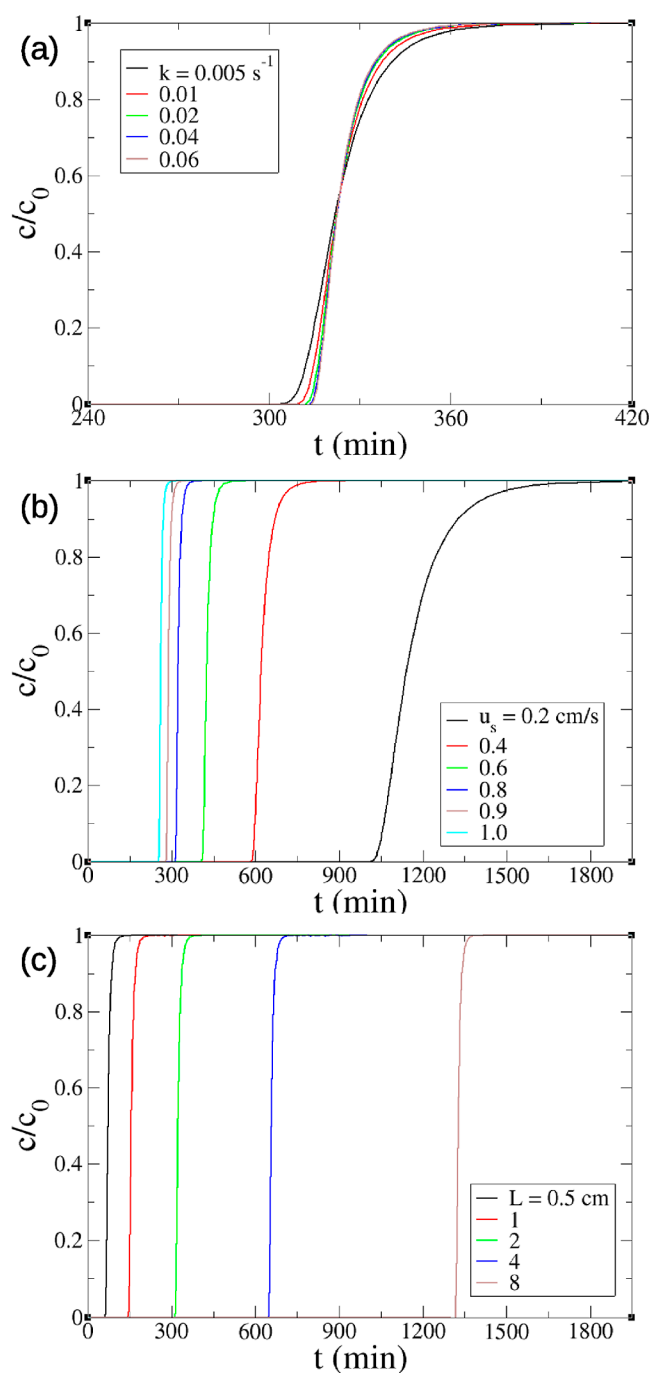


Figure 2. Breakthrough curves of the minimal model at DAC conditions (400 ppm of CO_2 , 25 °C, dry) with varying parameters: (a) uptake rate constant k ; (b) superficial velocity u_s ; and (c) column length L .

(Figure 2b) because a larger gas velocity or flow rate gives rise to a stronger convection contribution, causing a larger concentration difference and, hence, greater driving force. Figure 2c shows that the longer the packed column, the slower the breakthrough because the adsorbent bed has a larger adsorption capacity. The effects of Toth model parameters on the breakthrough curves are shown in Figure 3. The larger n_s , b , and t_T lead to a slower breakthrough due to the larger equilibrium adsorption q_e , since it takes more time for the packed column to get saturated as the equilibrium adsorption capacity increases.

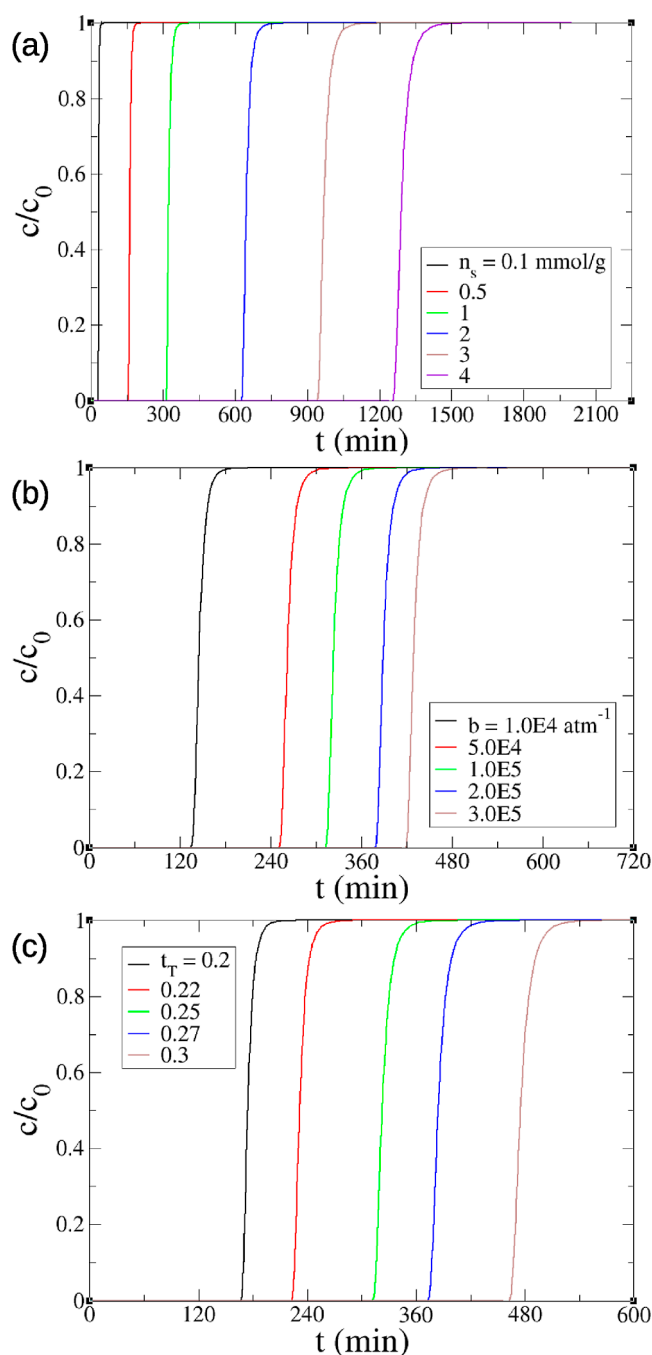


Figure 3. Effects of Toth model parameters on the breakthrough curves of the minimal model at DAC conditions (400 ppm of CO_2 , 25 °C, dry): (a) saturation adsorption n_s ; (b) adsorption affinity b ; and (c) Toth exponent t_T .

3.3. Humidity Effect. Water vapor is ubiquitous in ambient air, and the water concentration is significantly higher than the CO_2 concentration; therefore, including the H_2O component in the model mixture to properly consider cooperative CO_2 coadsorption with moisture is essential. The air was modeled as a ternary mixture of CO_2 , H_2O , and N_2 at a constant composition of 400 ppm of CO_2 , with a 40% relative humidity of H_2O and the remainder of N_2 . Note that N_2 was used to model all inert carrier components in air, which have low or negligible affinity to amine adsorbents. So, the LDF (eq 2) and minimum model (eq 10) were applied to both

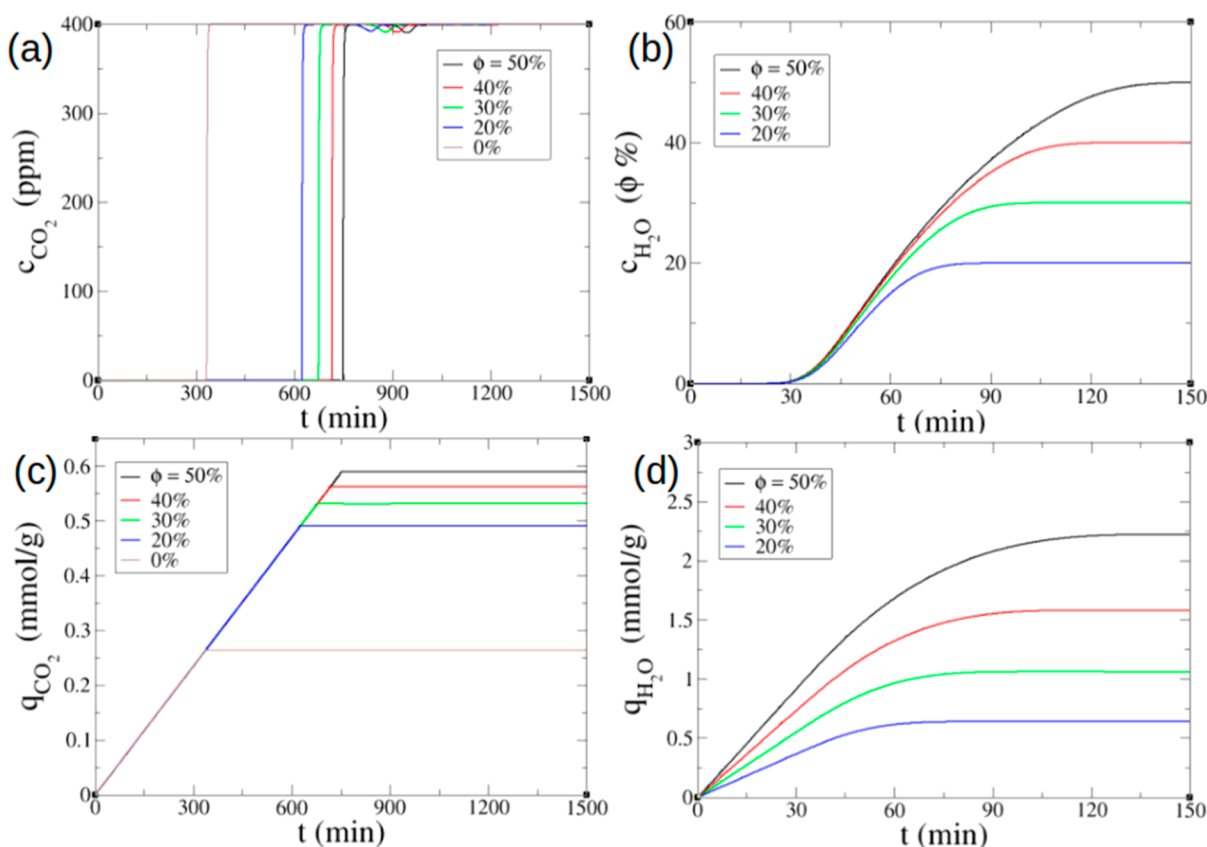


Figure 4. Effect of relative humidity (ϕ) on DAC (400 ppm of CO_2 , 25 °C) simulated with the minimal model for the packed bed: (a) CO_2 breakthrough curves; (b) H_2O breakthrough curves; (c) CO_2 dynamic adsorption capacity; and (d) H_2O dynamic adsorption capacity.

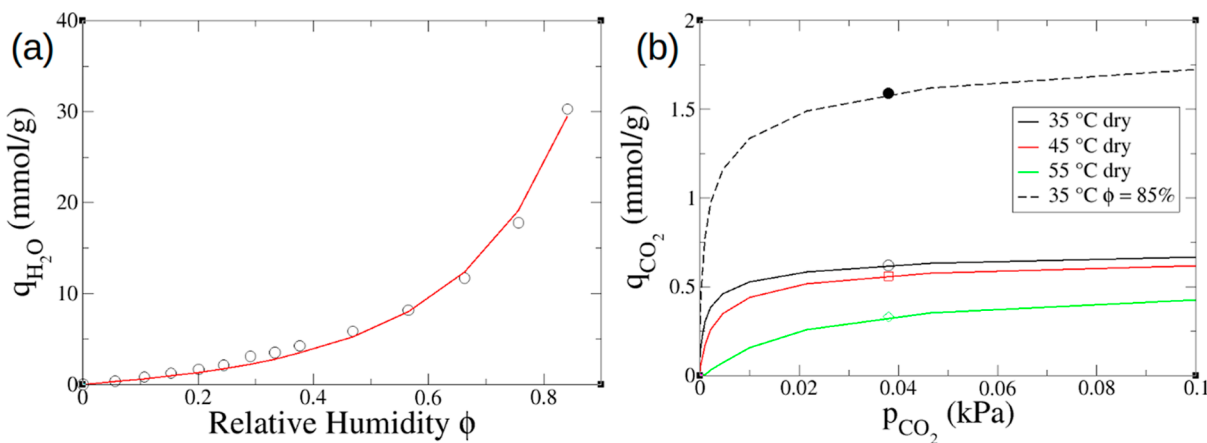


Figure 5. Application of our minimal model to the experimental data of DAC by PEI-impregnated silica fiber sorbents in humid conditions (380 ppm of CO_2): (a) fitted H_2O equilibrium uptake against relative humidity (line) in comparison with the experiment (symbol);⁴⁶ (b) fitted CO_2 isotherms (line) at different temperatures in comparison with the experiment (symbols).⁴⁶

CO_2 (c_1) and H_2O (c_2). To consider CO_2 coadsorption with moisture, a modified Toth model with enhanced adsorption affinity and saturated adsorption capacity was applied.³¹ When H_2O is present in the amine adsorbents, n_s and b in the Toth isotherm are updated accordingly as a function of temperature, CO_2 partial pressure, and adsorbed H_2O quantity: $n_s(q_2) = n_s(1/(1 - \gamma q_2))$ and $b(q_2) = b(1 + \beta q_2)$, where γ and β are positive parameters (Table 1). H_2O isotherm was described by the GAB model,⁴⁴ whose parameters are also shown in Table 1.

Figure 4 shows the effect of relative humidity (ϕ) on DAC. One can see that the CO_2 breakthrough curves shift to longer times when the humidity level increases (Figure 4a) due to the enhanced CO_2 adsorption capacity with moisture. The improved capacity in humid conditions, captured by the modified Toth isotherm with an empirical enhancement factor, is due to the opening of an additional reaction pathway which allows bicarbonate formation from water, amine group, and CO_2 .⁴⁵ In dry conditions, CO_2 breaks through the adsorption column around 330 min, while 630 min at $\phi = 20\%$ and 750 min at $\phi = 50\%$ (Figure 4a). In comparison, H_2O

concentration at the outlet reaches the feed saturated level much faster (Figure 4b): it hardly changes after 80 min at $\phi = 20\%$ and 140 min at $\phi = 50\%$. The dynamic adsorption capacity of the bed as a function of time can be used to estimate the breakthrough uptake given the breakthrough time, which was defined as the time corresponding to an arbitrary ratio, e.g., $c/c_0 = 0.1$, at the breakthrough curve. For instance, the breakthrough CO_2 uptake at $\phi = 20\%$ is about 0.49 mmol/g (Figure 4c) and H_2O uptake is around 0.62 mmol/g (Figure 4d).

3.4. Application of Our Minimal Model to a Recent DAC Experiment. We applied the minimal model to simulate a recent experiment where the authors studied temperature, flow rate, and humidity effects on breakthrough curves of PEI-impregnated silica fiber sorbents with a 380 ppm of CO_2 feed.⁴⁶ To begin with, the water isotherm data was fitted to the GAB model with $c_m = 26$ mmol/g, $c_G = 0.22$, and $K_{\text{ads}} = 0.84$ (Figure 5a). The pseudoequilibrium CO_2 capacities obtained from the breakthrough experiment were used to approximate the equilibrium data. The Toth model with the following parameters ($T_0 = 308$ K, $n_{s0} = 0.81$ mmol/g, $b_0 = 6.2 \times 10^3$ kPa⁻¹, $t_{T0} = 0.40$, $\Delta H_0 = 210$ kJ/mol, and $\chi = 6.6$, $\alpha = 10.8$) can well capture temperature dependence of the CO_2 isotherm. An empirical enhancement factor of $e^{1.11\phi}$ was found to describe improved CO_2 coadsorption with moisture; the resulting single (dry) and binary (humid) CO_2 isotherms present a realistic representation of experimental data (Figure 5b).

We further simulated the effects of temperatures and flow rates on the breakthrough curves and found excellent agreement between the modeling prediction and experiment (Figure 6). As temperature increases, the difference in breakthrough diminishes as a function of flow rate. Fitting the optimized uptake rate constants (0.04 min^{-1} at 35°C , 0.048 min^{-1} at 45°C , and 0.1 min^{-1} at 55°C) at the dry DAC conditions to the Arrhenius equation leads to an adsorption activation energy of $E_a = 38 \pm 14$ kJ/mol, which is in reasonable agreement with the measured activation energy of DAC by PEI-functionalized silica sorbent in the similar conditions (28.6 kJ/mol).²² It is generally believed that increasing the temperature improves CO_2 diffusion through the aggregated amine phase inside the pores of solid support, thereby facilitating CO_2 adsorption.²⁰ Yet, the favorable kinetics with increasing temperature is not significant enough to overcome the dominant thermodynamic trend; that is, increasing temperature lowers CO_2 uptake capacity. The overall effect explains the general trend of breakthrough and pseudoequilibrium uptakes observed in the breakthrough experiment. One can see from Table 3 that both the experimental breakthrough (q_b) and pseudoequilibrium (q_e) capacities decrease with temperature, while the breakthrough capacity (q_b) also decreases with the flow rate.⁴⁶ In higher flow rates, CO_2 molecules have less residence time in the packed bed to reach adsorption equilibrium, leading to sharper breakthrough curves and lower breakthrough capacities. Table 3 shows that the predicted breakthrough and pseudoequilibrium capacities agree well with the experimental values.

We further investigated the effect of humidity on the CO_2 breakthrough at 85% relative humidity. The predicted breakthrough curves are in good agreement with those from the experiment (Figure 7). A significant slowdown of the CO_2 adsorption kinetics was found. The CO_2 uptake rate constant

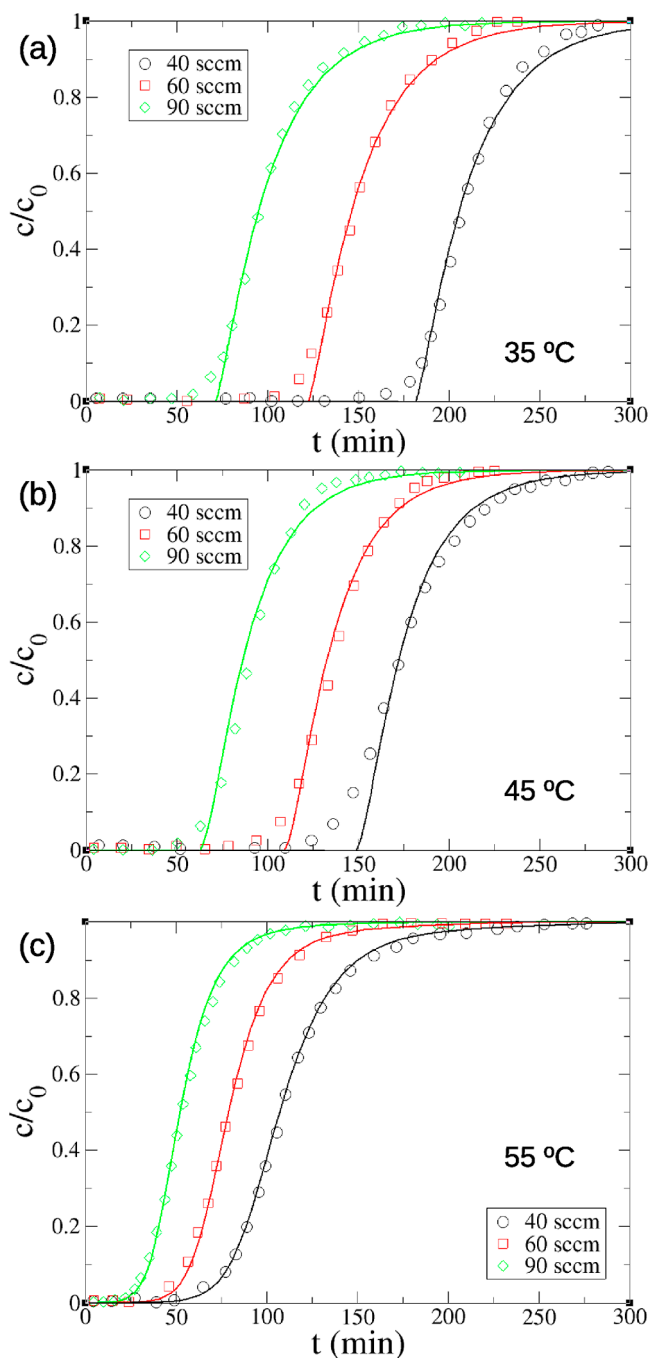
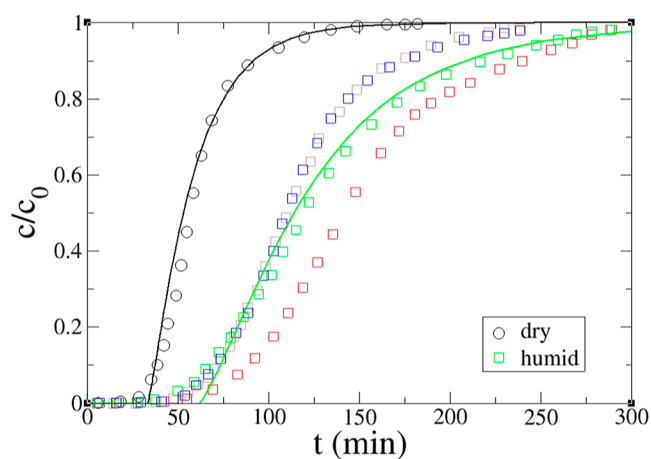


Figure 6. Comparison of dry DAC breakthrough curves (380 ppm of CO_2) between modeling prediction (line) and experiment⁴⁶ (symbol) at different feed flow rates and temperatures: (a) 35°C ; (b) 45°C ; and (c) 55°C . The optimized uptake rate constants are 0.04 min^{-1} at 35°C , 0.048 min^{-1} at 45°C , and 0.1 min^{-1} at 55°C (1 sccm = 1 cm^3 STP/min; STP is the standard condition of 0°C and 1 bar).

decreases from 0.04 min^{-1} in dry conditions to 0.02 min^{-1} in humid conditions, while the H_2O uptake rate constant is much faster at 0.36 min^{-1} . Improved CO_2 adsorption upon the addition of moisture is generally explained by the formation of additional carbonate/bicarbonate species and enhanced amine accessibility due to facile CO_2 diffusion.¹⁰ Our modeling result suggests that such improvements do not necessarily lead to faster CO_2 adsorption kinetics. The slower kinetics of CO_2 adsorption under humid conditions could be due to slower gas diffusion in the sorbent or a slower reaction with the amines in

Table 3. Comparison of Prediction and Experiment⁴⁶ of Breakthrough (q_b) and Pseudoequilibrium (q_e) Capacities at 5% c/c_0 and 95% c/c_0 , Respectively, at DAC Conditions (380 ppm of CO_2 , Dry)

	q_b (mmol/g): prediction/experiment			q_e (mmol/g): prediction/experiment		
	40sccm	60sccm	90sccm	40sccm	60sccm	90sccm
35 °C	0.53/0.51	0.50/0.48	0.44/0.44	0.62/0.61	0.62/0.61	0.62/0.62
45 °C	0.47/0.41	0.43/0.37	0.39/0.36	0.56/0.55	0.56/0.56	0.56/0.56
55 °C	0.21/0.19	0.19/0.13	0.16/0.08	0.33/0.32	0.33/0.33	0.33/0.33

**Figure 7.** Comparison of modeling prediction (line) and experiment (symbol) of dry and humid 380 ppm of CO_2 breakthrough curves at 35 °C and 200 sccm. In the experiment,⁴⁶ multiple independent runs (denoted by open squares of different colors) of humid CO_2 were performed.

the presence of water. Our current model lumped those factors into one kinetic parameter, the CO_2 uptake rate constant (k), and could not differentiate their contributions. In terms of capacity, the predicted breakthrough and pseudoequilibrium capacities are in good agreement with the experiment in both dry and humid conditions (Table 4). The overall effects of various conditions are summarized in Table 5.

Table 4. Comparison of Prediction and Experiment⁴⁶ of Breakthrough (q_b) and Pseudoequilibrium (q_e) Capacities at 5% c/c_0 and 95% c/c_0 , Respectively, at Dry and Humid Conditions at 35 °C and 200 sccm (380 ppm of CO_2)

	q_b (mmol/g): prediction/experiment	q_e (mmol/g): prediction/experiment
dry	0.37/0.36	0.62/0.62
humid	0.80/0.64	1.60/1.60

3.5. Limitations of Our Model. Despite its simplicity and utility, the minimal model that we have developed here has limitations due to the assumptions made. We summarize here the assumptions and limitations: (1) plug flow, constant gas superficial velocity, no axial dispersion; (2) isotherm model, in other words, heat transfer has an insignificant effect on the breakthrough curve in the benchtop scale (nonisotherm model with energy balance is necessary for DAC cycle process modeling and techno-economic analysis); (3) one-dimensional model, so gas concentration gradients only exist in the axial direction; (4) gas phase behaves as ideal gas; (5) negligible pressure drop due to the short column length and relatively low flow rate (pressure drop can be described by the Ergun equation if needed); (6) only CO_2 and H_2O adsorption are

Table 5. Summary of the Effects of the Varying Conditions on the DAC Breakthrough Curves from Our Minimal Kinetic Model

conditions	effects
increasing uptake rate constant	steeper breakthrough
increasing flow rate	faster breakthrough, lower breakthrough capacity
increasing fixed bed length	slower breakthrough, higher breakthrough capacity
increasing equilibrium capacity	slower breakthrough, higher breakthrough capacity
increasing temperature	lower equilibrium capacity, faster adsorption kinetics, faster breakthrough, lower breakthrough capacity
humid CO_2	larger equilibrium capacity, slower and broader breakthrough, higher breakthrough capacity

considered, and all others are treated as nonadsorbing components; and (7) LDF adsorption kinetics model (more complex mass transfer resistance model could be applied to describe different mass transfer resistances from the gas phase to the solid sorbent phase).

4. CONCLUSIONS

CO_2 under dry and humid DAC conditions was kinetically simulated using a minimal mathematical model. Parameters related to process operation, isotherm, and adsorption kinetics were varied to study the general effect on breakthrough curves. The larger uptake rate leads to a steeper breakthrough. Either the longer column or the slower gas velocity gives rise to a slower breakthrough. The larger equilibrium adsorption arising from the larger saturated adsorption or adsorption affinity or Toth exponent also leads to a slower breakthrough. The humidity effect on breakthrough curves and bed adsorption capacity were studied with the help of a modified Toth isotherm to properly consider enhanced CO_2 coadsorption with moisture. Increasing relative humidity shifts the breakthrough curve to a longer time with a higher adsorption capacity of the bed. The implemented methodology was successfully applied to model a recent DAC experiment of supported amine sorbent under both dry and humid conditions. Temperature and flow rate effects on breakthrough were quantitatively captured. The CO_2 uptake rate constantly increases with temperature. Moisture broadens the breakthrough curve with a significantly improved uptake capacity but slower CO_2 adsorption kinetics. The minimal model developed in this work will be useful in understanding the thermodynamics and kinetics of the DAC of CO_2 on supported amine and other chemisorption systems.

AUTHOR INFORMATION

Corresponding Author

De-en Jiang — Department of Chemical and Biomolecular Engineering, Vanderbilt University, Nashville, Tennessee

37235, United States; orcid.org/0000-0001-5167-0731;
Email: de-en.jiang@vanderbilt.edu

Authors

Hongjun Liu – Department of Chemical and Biomolecular Engineering, Vanderbilt University, Nashville, Tennessee 37235, United States

Hongfei Lin – The Gene and Linda Voiland School of Chemical Engineering and Bioengineering, Washington State University, Pullman, Washington 99164, United States; orcid.org/0000-0001-8198-1493

Sheng Dai – Chemical Sciences Division, Oak Ridge National Laboratory, Oak Ridge, Tennessee 37831, United States; Department of Chemistry, University of Tennessee, Knoxville, Tennessee 37996, United States; orcid.org/0000-0002-8046-3931

Complete contact information is available at:
<https://pubs.acs.org/10.1021/acs.iecr.3c04535>

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This work was supported by the DOE Office of Basic Energy Sciences under grant number DE-SC0022273.

REFERENCES

- (1) Sanz-Perez, E. S.; Murdock, C. R.; Didas, S. A.; Jones, C. W. Direct Capture of CO₂ from Ambient Air. *Chem. Rev.* **2016**, *116*, 11840–11876.
- (2) Kumar, D. R.; Rosu, C.; Sujun, A.; Sakwa-Novak, M. A.; Ping, E. W.; Jones, C. W. Alkyl-Aryl Amine-Rich Molecules for CO₂ Removal via Direct Air Capture. *ACS Sustainable Chem. Eng.* **2020**, *8*, 10971–10982.
- (3) Shi, X.; Xiao, H.; Azarabadi, H.; Song, J.; Wu, X.; Chen, X.; Lackner, K. S. Sorbents for the Direct Capture of CO₂ from Ambient Air. *Angew. Chem., Int. Ed.* **2020**, *59*, 6984–7006.
- (4) Xu, X.; Song, C.; Andresen, J. M.; Miller, B. G.; Scaroni, A. W. Novel Polyethylenimine-Modified Mesoporous Molecular Sieve of MCM-41 Type as High-Capacity Adsorbent for CO₂ Capture. *Energy Fuels* **2002**, *16*, 1463–1469.
- (5) Belmabkhout, Y.; Sayari, A. Effect of pore expansion and amine functionalization of mesoporous silica on CO₂ adsorption over a wide range of conditions. *Adsorption* **2009**, *15*, 318–328.
- (6) Bollini, P.; Didas, S. A.; Jones, C. W. Amine-oxide hybrid materials for acid gas separations. *J. Mater. Chem.* **2011**, *21*, 15100–15120.
- (7) Chaikittisilp, W.; Kim, H.-J.; Jones, C. W. Mesoporous Alumina-Supported Amines as Potential Steam-Stable Adsorbents for Capturing CO₂ from Simulated Flue Gas and Ambient Air. *Energy Fuels* **2011**, *25*, 5528–5537.
- (8) Choi, S.; Gray, M. L.; Jones, C. W. Amine-Tethered Solid Adsorbents Coupling High Adsorption Capacity and Regenerability for CO₂ Capture From Ambient Air. *ChemSusChem* **2011**, *4*, 628–635.
- (9) Hamdy, L. B.; Goel, C.; Rudd, J. A.; Barron, A. R.; Andreoli, E. The application of amine-based materials for carbon capture and utilisation: an overarching view. *Mater. Adv.* **2021**, *2*, 5843–5880.
- (10) Kolle, J. M.; Fayaz, M.; Sayari, A. Understanding the Effect of Water on CO₂ Adsorption. *Chem. Rev.* **2021**, *121*, 7280–7345.
- (11) Didas, S. A.; Choi, S.; Chaikittisilp, W.; Jones, C. W. Amine-Oxide Hybrid Materials for CO₂ Capture from Ambient Air. *Acc. Chem. Res.* **2015**, *48*, 2680–2687.
- (12) Pang, S. H.; Jue, M. L.; Leisen, J.; Jones, C. W.; Lively, R. P. PIM-1 as a Solution-Processable “Molecular Basket” for CO₂ Capture from Dilute Sources. *ACS Macro Lett.* **2015**, *4*, 1415–1419.
- (13) Chai, S.-H.; Liu, Z.-M.; Huang, K.; Tan, S.; Dai, S. Amine Functionalization of Microsized and Nanosized Mesoporous Carbons for Carbon Dioxide Capture. *Ind. Eng. Chem. Res.* **2016**, *55*, 7355–7361.
- (14) Lyu, H.; Chen, O. I.-F.; Hanikel, N.; Hossain, M. I.; Flaig, R. W.; Pei, X.; Amin, A.; Doherty, M. D.; Impastato, R. K.; Glover, G.; Moore, D. R.; Yaghi, O. M. Carbon Dioxide Capture Chemistry of Amino Acid Functionalized Metal Organic Frameworks in Humid Flue Gas. *J. Am. Chem. Soc.* **2022**, *144*, 2387–2396.
- (15) Danckwerts, P. The reaction of CO₂ with ethanolamines. *Chem. Eng. Sci.* **1979**, *34*, 443–446.
- (16) Yang, Z.-Z.; He, L.-N.; Zhao, Y.-N.; Li, B.; Yu, B. CO₂ capture and activation by superbases/polyethylene glycol and its subsequent conversion. *Energy Environ. Sci.* **2011**, *4*, 3971–3975.
- (17) Priyadarshini, P.; Rim, G.; Rosu, C.; Song, M.; Jones, C. W. Direct Air Capture of CO₂ Using Amine/Alumina Sorbents at Cold Temperature. *ACS Environ. Au* **2023**, *3*, 295–307.
- (18) Zeeshan, M.; Kidder, M. K.; Pentzer, E.; Getman, R. B.; Gurkan, B. Direct air capture of CO₂: from insights into the current and emerging approaches to future opportunities. *Front. Sustainability* **2023**, *4*, 1167713.
- (19) Serna-Guerrero, R.; Belmabkhout, Y.; Sayari, A. Modeling CO₂ adsorption on amine-functionalized mesoporous silica: 1. A semi-empirical equilibrium model. *Chem. Eng. J.* **2010**, *161*, 173–181.
- (20) Serna-Guerrero, R.; Sayari, A. Modeling adsorption of CO₂ on amine-functionalized mesoporous silica. 2: Kinetics and breakthrough curves. *Chem. Eng. J.* **2010**, *161*, 182–190.
- (21) Bollini, P.; Brunelli, N. A.; Didas, S. A.; Jones, C. W. Dynamics of CO₂ Adsorption on Amine Adsorbents. 1. Impact of Heat Effects. *Ind. Eng. Chem. Res.* **2012**, *51*, 15145–15152.
- (22) Jung, W.; Park, J.; Lee, K. S. Kinetic modeling of CO₂ adsorption on an amine-functionalized solid sorbent. *Chem. Eng. Sci.* **2018**, *177*, 122–131.
- (23) Gebald, C.; Wurzbacher, J. A.; Borgschulte, A.; Zimmermann, T.; Steinfeld, A. Single-Component and Binary CO₂ and H₂O Adsorption of Amine-Functionalized Cellulose. *Environ. Sci. Technol.* **2014**, *48*, 2497–2504.
- (24) Sayari, A.; Liu, Q.; Mishra, P. Enhanced Adsorption Efficiency through Materials Design for Direct Air Capture over Supported Polyethylenimine. *ChemSusChem* **2016**, *9*, 2796–2803.
- (25) Elfving, J.; Bajamundi, C.; Kauppinen, J.; Sainio, T. Modelling of equilibrium working capacity of PSA, TSA and TVSA processes for CO₂ adsorption under direct air capture conditions. *J. CO₂ Util.* **2017**, *22*, 270–277.
- (26) Moitra, D.; Mokhtari-Nori, N.; Siniard, K. M.; Qiu, L.; Fan, J.; Dong, Z.; Hu, W.; Liu, H.; Jiang, D.-e.; Lin, H.; Hu, J.; Li, M.; Yang, Z.; Dai, S. High-Performance CO₂ Capture from Air by Harnessing the Power of CaO- and Superbase-Ionic-Liquid-Engineered Sorbents. *ChemSusChem* **2023**, *16*, No. e202300808.
- (27) Qiu, L.; Peng, L.; Moitra, D.; Liu, H.; Fu, Y.; Dong, Z.; Hu, W.; Lei, M.; Jiang, D. E.; Lin, H.; Hu, J.; McGarry, K. A.; Popovs, I.; Li, M.; Ivanov, A. S.; Yang, Z.; Dai, S. Harnessing the Hybridization of a Metal-Organic Framework and Superbase-Derived Ionic Liquid for High-Performance Direct Air Capture of CO₂. *Small* **2023**, *19*, 2302708.
- (28) Wang, J.; Wang, M.; Li, W.; Qiao, W.; Long, D.; Ling, L. Application of polyethylenimine-impregnated solid adsorbents for direct capture of low-concentration CO₂. *AIChE J.* **2015**, *61*, 972–980.
- (29) Xu, X.; Myers, M. B.; Versteeg, F. G.; Pejčić, B.; Heath, C.; Wood, C. D. Direct air capture (DAC) of CO₂ using polyethylenimine (PEI) “snow”: a scalable strategy. *Chem. Commun.* **2020**, *56*, 7151–7154.
- (30) Jung, W.; Lee, K. S. Isotherm and kinetics modeling of simultaneous CO₂ and H₂O adsorption on an amine-functionalized solid sorbent. *J. Nat. Gas Sci. Eng.* **2020**, *84*, 103489.
- (31) Stampi-Bombelli, V.; van der Spek, M.; Mazzotti, M. Analysis of direct capture of CO₂ from ambient air via steam-assisted temperature-vacuum swing adsorption. *Adsorption* **2020**, *26*, 1183–1197.

(32) Elfving, J.; Sainio, T. Kinetic approach to modelling CO₂ adsorption from humid air using amine-functionalized resin: Equilibrium isotherms and column dynamics. *Chem. Eng. Sci.* **2021**, *246*, 116885.

(33) Jahandar Lashaki, M.; Khiavi, S.; Sayari, A. Stability of amine-functionalized CO₂ adsorbents: a multifaceted puzzle. *Chem. Soc. Rev.* **2019**, *48*, 3320–3405.

(34) Moon, H. J.; Carrillo, J.-M.; Leisen, J.; Sumpter, B. G.; Osti, N. C.; Tyagi, M.; Jones, C. W. Understanding the Impacts of Support-Polymer Interactions on the Dynamics of Poly(ethyleneimine) Confined in Mesoporous SBA-15. *J. Am. Chem. Soc.* **2022**, *144*, 11664–11675.

(35) Short, G. N.; Burentugs, E.; Proaño, L.; Moon, H. J.; Rim, G.; Nezam, I.; Korde, A.; Nair, S.; Jones, C. W. Single-Walled Zeolitic Nanotubes: Advantaged Supports for Poly(ethylenimine) in CO₂ Separation from Simulated Air and Flue Gas. *JACS Au* **2023**, *3*, 62–69.

(36) Buijs, W.; de Flart, S. Direct Air Capture of CO₂ with an Amine Resin: A Molecular Modeling Study of the CO₂ Capturing Process. *Ind. Eng. Chem. Res.* **2017**, *56*, 12297–12304.

(37) Zhang, H.; Yang, L.-M.; Ganz, E. Adsorption Properties and Microscopic Mechanism of CO₂ Capture in 1,1-Dimethyl-1,2-ethylenediamine-Grafted Metal-Organic Frameworks. *ACS Appl. Mater. Interfaces* **2020**, *12*, 18533–18540.

(38) Orlov, A. A.; Valtz, A.; Coquelet, C.; Rozanska, X.; Wimmer, E.; Marcou, G.; Horvath, D.; Poulain, B.; Varnek, A.; de Meyer, F. Computational screening methodology identifies effective solvents for CO₂ capture. *Commun. Chem.* **2022**, *5*, 37.

(39) Chen, J.; Moon, H. J.; Kim, K. I.; Choi, J. I.; Narayanan, P.; Sakwa-Novak, M. A.; Jones, C. W.; Jang, S. S. Distribution and Transport of CO₂ in Hyperbranched Poly(ethylenimine)-Loaded MCM-41: A Molecular Dynamics Simulation Approach. *ACS Appl. Mater. Interfaces* **2023**, *15*, 43678–43690.

(40) Young, J.; García-Díez, E.; García, S.; van der Spek, M. The impact of binary water-CO₂ isotherm models on the optimal performance of sorbent-based direct air capture processes. *Energy Environ. Sci.* **2021**, *14*, 5377–5394.

(41) Shafeeyan, M. S.; Wan Daud, W. M. A.; Shamiri, A. A review of mathematical modeling of fixed-bed columns for carbon dioxide adsorption. *Chem. Eng. Res. Des.* **2014**, *92*, 961–988.

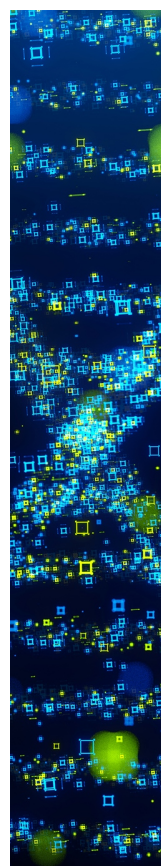
(42) Sircar, S.; Hufton, J. Why Does the Linear Driving Force Model for Adsorption Kinetics Work? *Adsorption* **2000**, *6*, 137–147.

(43) Bollini, P.; Brunelli, N. A.; Didas, S. A.; Jones, C. W. Dynamics of CO₂ Adsorption on Amine Adsorbents. 2. Insights Into Adsorbent Design. *Ind. Eng. Chem. Res.* **2012**, *51*, 15153–15162.

(44) Guggenheim, E. A. *Applications of Statistical Mechanics*; Clarendon Press: Oxford, U.K., 1996.

(45) Kortunov, P. V.; Siskin, M.; Paccagnini, M. L.; Thomann, H. CO₂ Reaction Mechanisms with Hindered Alkanolamines: Control and Promotion of Reaction Pathways. *Energy Fuels* **2016**, *30*, 1223–1236.

(46) Sujan, A. R.; Pang, S. H.; Zhu, G.; Jones, C. W.; Lively, R. P. Direct CO₂ Capture from Air using Poly(ethylenimine)-Loaded Polymer/Silica Fiber Sorbents. *ACS Sustainable Chem. Eng.* **2019**, *7*, 5264–5273.



CAS BIOFINDER DISCOVERY PLATFORM™

STOP DIGGING THROUGH DATA —START MAKING DISCOVERIES

CAS BioFinder helps you find the
right biological insights in seconds

Start your search



A Division of the
American Chemical Society